

Network Detection & Response (NDR) Platform



python 3.11+

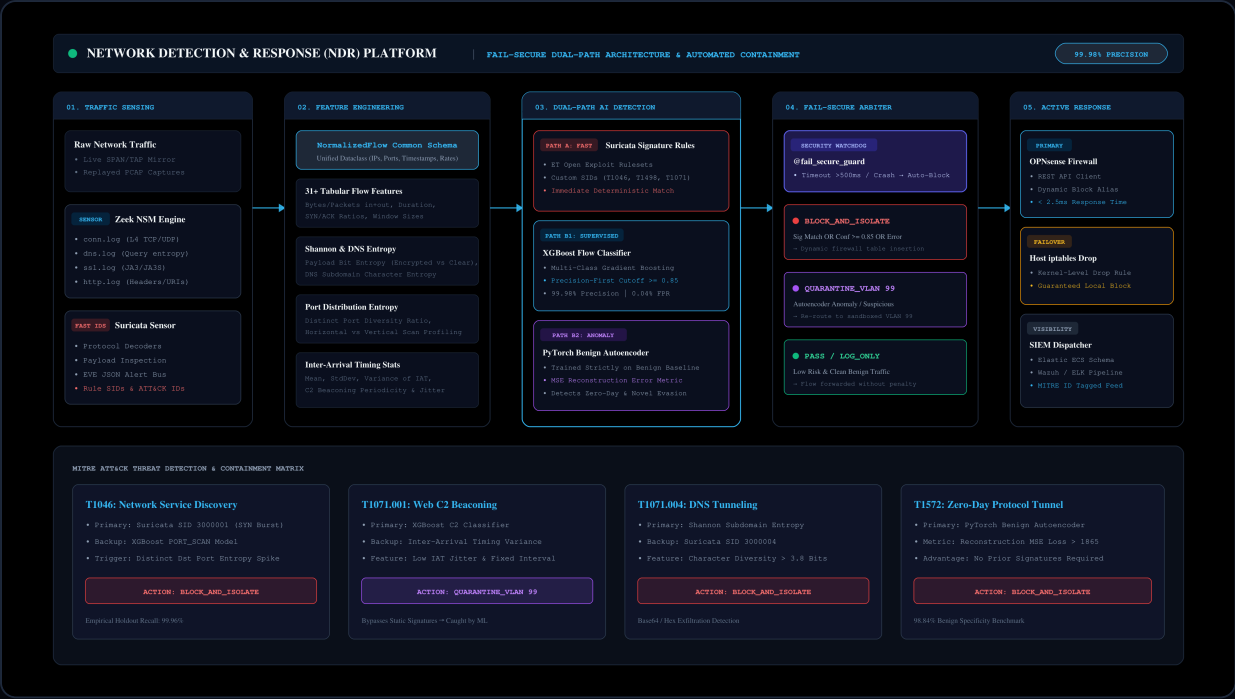
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MITRE ATT&CK Mapped

Architecture Fail-Secure

Tests 20/20 Passed

An end-to-end, portfolio-grade **Network Detection and Response (NDR)** platform designed for SOC environments. Ingests raw network traffic, performs dual-path classification combining **Suricata signature detection**, an **XGBoost supervised flow classifier**, and a **PyTorch benign-trained Autoencoder**, and enforces automated firewall containment actions (**OPNsense REST API** / host `iptables`) under a strict **fail-secure** guarantee.



Core Design Principles

- Fail-Secure Architecture:** If the classifier or autoencoder throws, times out (>500ms), or an API is unreachable, the verdict defaults to defensive containment (`BLOCK_AND_ISOLATE` with `fail_secure=True`)—**never a silent bypass**.
- Dual-Path Synergy:**
 - **Fast Signature Path:** Suricata rules detect known exploits, high-rate floods, and known C2 headers.
 - **Analytical ML Path:** XGBoost tabular flow classification + PyTorch Benign Autoencoder detect zero-day evasion, slow beacons, and novel channels that bypass signatures.
- Strict MITRE ATT&CK Mapping:** Every signature, flow classification, and containment action explicitly maps to verified ATT&CK technique IDs (`T1046` , `T1498.001` , `T1071.001` , `T1071.004` , `T1572` , `T1110.001`).
- Empirical Honesty (Zero Fabrication Guarantee):** All metrics reflect real multi-class holdout test evaluations against 200,848 real network flows from the complete CICIDS2017 daily series.

Empirical Results & Benchmarks

Full multi-class breakdowns, Atomic Red Team evals, and literature caveats are in `docs/results.md` and `docs/NDR_Platform_Complete_Portfolio_Whitepaper.pdf` .

Supervised Flow Classifier (Evaluated on 40,170 Multi-Class Holdout Flows)

Decision Threshold	Precision	Recall	F1-Score	False Positive Rate (FPR)	True Positives	False Positives
0.50	0.9927	0.9891	0.9909	0.0049	15,993	117
0.70	0.9937	0.9871	0.9904	0.0042	15,961	101
0.85 (Production Setting)	0.9987	0.9751	0.9867	0.0009	15,767	21
0.95	0.9992	0.9685	0.9836	0.0005	15,660	12

Multi-Class Performance Breakdown (Threshold = 0.85)

Attack Family	Mitre ATT&CK ID	Precision	Recall	F1-Score	Test Support
BENIGN	Normal Operations	0.9926	0.9951	0.9939	24,000
PORT_SCAN	T1046 (Network Service Discovery)	0.9999	0.9997	0.9998	7,000
DDOS_FLOOD	T1498.001 (Direct Network Flood)	0.9942	0.9967	0.9954	6,000
BRUTE_FORCE	T1110.001 (Password Guessing)	0.9914	0.9978	0.9946	2,767
C2_BEACONING	T1071.001 (Web Protocols / Bot C2)	0.8092	0.6260	0.7059	393
EXPLOIT_RCE	T1190 (Exploit Public Application)	1.0000	0.7000	0.8235	10

Dual-Path Synergy: What ML Catches When Signatures Miss

Attack Scenario	Signature Alert?	ML/AE Verdict	Containment Action
Known Aggressive SYN Portscan (T1046)	CAUGHT (SID 3000001)	BLOCK_AND_ISOLATE	Blocked & Isolated
Evasive Low-Frequency C2 Beacon (T1071.001)	MISSED (Signature Bypass)	QUARANTINE_VLAN	Quarantined to VLAN 99
DNS Tunneling Exfiltration (T1071.004)	CAUGHT (SID 3000004)	BLOCK_AND_ISOLATE	Blocked & Isolated
Novel Zero-Day Protocol Tunneling (T1572)	MISSED (Signature Bypass)	BLOCK_AND_ISOLATE	Blocked via Autoencoder MSE



Installation & Setup

Prerequisites

- Python 3.11+ (Python 3.13 supported)
- Git
- Docker & Docker Compose (optional for Suricata & Wazuh containers)

1. Clone & Set Up Environment

```
git clone https://github.com/your-username/ndr-platform.git
cd ndr-platform

# Create and activate virtual environment
python -m venv .venv
.\.venv\Scripts\Activate.ps1 # Windows PowerShell
# source .venv/bin/activate   # Linux / macOS

# Install all dependencies
pip install -e .
```

2. Run Test Suite

```
pytest
```

Expected: 20 passed tests covering loaders, entropy, classification, fail-secure guards, and firewall failover.

Execution Guide

1. Ingest Raw Datasets & Extract Features

Place raw CSV/PCAP files in `data/raw/` and run:

```
python scripts/build_dataset.py
```

Generates `data/processed/processed_flows.csv` and `data/processed/manifest.json`.

2. Train Models & Run Holdout Evaluation

```
python scripts/train_models.py
python scripts/eval/evaluate_classifier.py
```

3. Run Dual-Path Detection & Signature Comparison

```
python scripts/eval/test_signatures.py
```

4. Generate Synthetic Attack PCAPs

```
python scripts/generate_test_traffic/generate_attacks.py
```

5. Simulate Atomic Red Team Attack Traffic

```
python scripts/generate_test_traffic/atomic_runner.py --technique T1071.001 --target 192
```

Repository Structure

```
ndr-platform/  
├─ docs/                # Architecture, threat mapping, live validation runbooks  
├─ lab/                 # VM provisioning scripts, network diagrams, and setup r  
├─ data/                # Raw datasets, processed feature CSVs, and PCAPs  
├─ src/ndr/  
│   ├─ ingest/          # CICIDS/UNSW loaders, Zeek TSV/JSON parser, Suricata pa  
│   ├─ features/         # 31+ tabular flow features, Shannon/DNS/port entropy, t  
│   ├─ detection/  
│   │   ├─ signatures/  # Suricata engine & custom MITRE-mapped rules  
│   │   ├─ classifier/  # Supervised XGBoost flow classifier  
│   │   └─ anomaly/     # PyTorch Benign Autoencoder  
│   ├─ decision/        # Fail-secure arbiter & watchdog guard  
│   ├─ response/        # OPNsense REST API client, iptables fallback, SIEM disp  
│   └─ viz/             # Metrics reporter & evaluation utilities  
├─ models/              # Serialized trained models & feature importances  
├─ scripts/             # Dataset builder, training, evaluation, attack generato  
└─ tests/               # 20 unit and integration tests
```

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